

Entity Resolution with FAISS + Splink

One system, three conversations

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How to use this deck

Part	Audience	Focus	After the talk
1	C-suite	Value, risk, cost	Decision / review
2	Engineering leadership	Architecture, design, roadmap	Carry it forward
3	Line engineers	Orientation, stack, hands-on	Go hands-on

Line engineers: this orients you. The whitepaper (`docs/entity_resolution_whitepaper.pdf`) is the source of record; the numbers here come from it.

Part 1 · Executive Summary

The 60-second version

The problem

Organisations hold **many records for the same real person**, and they disagree.

- Names differ (e.g., `R. Martinez` vs `Robert Martinez`).
- Addresses change; emails and dates go missing (**30% missing** here).
- Duplicates fragment CRM, risk, and reporting.

Wrong answers are expensive: linking two *different* people is usually far worse than missing two records of the same person.

The solution: two stages

Fast retrieval + explainable matching.

```
Query → [embed] → [FAISS top-k]
      → [Splink linkage] → Match / No match
```

- **Blocking** (FAISS): cheaply finds the few most likely matches.
- **Linkage** (Splink): weighs the evidence per field and returns a probability, not just yes/no.

What it delivers

- **99.5% F1** on the synthetic test corpus (15,000 labelled queries).
- **~7.5 ms per query** — sub-millisecond blocking, probabilistic verdict on top.
- **Persisted and reusable**: the index reloads without re-embedding the population.
- **Defensible numbers**: every result reproduces by re-running the code.

Set the risk dial, not a magic number

Splink returns a **probability**; a pair is called a match when that probability is at or above a threshold τ . One number encodes the business trade-off.

- **Raise** τ → fewer false positives (fewer *wrong* links) at slightly more false negatives.
- **Lower** τ → catch more true matches, at more false positives.

The cost-based rule:

$$\tau^* = C_{FP} / (C_{FP} + C_{FN})$$

If a **false link** (FP) is 20× more costly than a **missed link** (FN), then

$\tau^* = 20/21 \approx 0.95$ — the model is tuned to the cost of each mistake.

One parameter lets a risk-averse line (fraud/AML) and a recall-first line (marketing) share the same model, each tuned to its own economics.

Cost and scale

- The 50,000-record index is **~87 MB** today.
- Capacity is predictable from RAM;

RAM	Recommended capacity
16 GB	~6.9 M records
32 GB	~13.8 M records
128 GB	~55.1 M records

Around **7–14 M records** (or when you need replicated, multi-region durability) buy a **vector database**; before that the in-process index is the lean choice.

The honest caveats

- Results are measured on **synthetic data**; production needs a **labelled sample of true duplicates** to calibrate and validate.
- **Retraining the match model without also resetting the threshold backfires.** The shipped defaults work best until proper joint calibration. Don't let a team "just retune" today.
- Volume must clear our capacity thresholds before any external-infra spend.

The ask

- Sponsor a short **proof-on-real-data** phase: a representative, labelled sample of the population we must link (with access/privacy).
- We then commit to an F1 on *that* data, not the synthetic one.

Value in one line: near-perfect, explainable identity resolution at ~7.5 ms, with a priceable path to scale.

Part 2 · Engineering Leadership

Architecture, measurements, where to invest

Architecture at a glance

Query $\rightarrow$ MiniLM embed $\rightarrow$ FAISS top-k $\rightarrow$ Splink $\rightarrow$ $p(M \mid \gamma)$
▲
name / DOB / email / address comparisons, τ threshold

- Embeddings are **L2-normalized**, so FAISS inner product = cosine.
- Splink = **Fellegi–Sunter**: per-field weight $w = \log_2(m/u)$, combined with a match prior, compared to τ .
- A store abstraction lets us swap the in-memory index for an **external vector DB** later without changing blocking or linkage.

Where performance comes from

Blocking recall is the ceiling; Splink is where F1 is won or lost.

- Top-20 blocking recall: **99.88%** (default), **99.95%** (compact serialization).
- End-to-end recall: **99.41%** → the ~0.5% gap is lost in *linkage*, not blocking.
- A "perfect" blocker buys only **~0.2 more F1 points** (99.46 → ~99.69% ceiling).

Design implication: improving the embedding/blocking engine is **not** the lever today; linkage configuration is.

The measured levers

Lever	Effect	Verdict
Threshold τ	0.85→0.95 lifts F1 99.46→99.64%	Yes — cheap precision knob
Serialization	compact: recall 99.95%, F1 99.73%	Yes — small but real
Blocking size k	20→100: +recall, 2.4× Splink time	Diminishing returns
Weaken address	never beat full-strength	No (this data)
Retrain m/u	untrained F1 98.1% vs supervised 94.1%	Keep defaults

τ is the cost dial (engineering view)

Splink's decision rule: classify as *match* when $P(M|\gamma) \geq \tau$ (range $\sim[0.5, 0.999]$).

- **Raise** τ $\rightarrow$ fewer false positives, at more false negatives.
- **Lower** τ $\rightarrow$ more true matches, at more false positives.

Decision-theoretic setting: $\tau^* = C_{FP} / (C_{FP} + C_{FN})$.

- Measured: τ 0.85 $\rightarrow$ 0.95 lifted F1 99.46% $\rightarrow$ 99.64% (precision 99.51 $\rightarrow$ 99.94%, FP 49 $\rightarrow$ 6); recall slipped only 99.41 $\rightarrow$ 99.34%.
- With recall headroom, raising τ is the cheapest precision lever — the mechanism to encode FP/FN business costs. (Methods: whitepaper **Appendix — Deriving τ** .)

The calibration paradox

Retraining the match model (m/u) made results **worse**, not better.

1. **Prior coupling (dominant):** EM's fitted prior (0.0071 vs 0.0001) shifts all scores up → mass false positives. Pinning the prior raises EM's F1 from ~0.83 to ~0.95.
2. **A genuine residual:** even with the prior pinned, trained m/u stay below defaults (supervised 94.1% vs untrained 98.1%; EM optimum 95.5%) by over-weighting partial matches.

EM training generates candidate pairs by exact-equality blocking on two keys: `first_name` and `date_of_birth` (`block_on("first_name")` , `block_on("date_of_birth")`).

Rule: the default config is a coherent bundle (weights + prior + τ). Change any part and re-validate **all three** on held-out data.

The fix is a recipe, not a guess: **anchor the prior** (default or blocking-adjusted), fit m/u with the prior pinned, then **tune τ jointly**, with a score-shift guard. Full algorithm:

Appendix — Tuning the Match Prior.

See paper §8.1 and the joint $\tau \times \text{prior}$ table.

Capacity and when to buy infra

- **Memory is the binding constraint** (~1,746 bytes/record).
- Move to an **external vector DB** around **7–14 M records**, or when you need shared/multi-region **durability and replication**, or beyond **~50 M** as a rule of thumb.
- The persisted index reloads **without re-embedding**, so redeploy and multi-consumer serving are cheap.

Operational readiness

- Persisted, reload-friendly store with `update` / `delete` ; external-store interface is the vector-DB swap point.
- **Reproducible**: every experiment maps to a script; scripts accept `--input` .

Roadmap

1. Invest in **linkage calibration** (labelled pairs + joint τ /prior tuning).
2. Do **not** spend on blocking-engine research at this scale. (Future: canopy/embedding-blocked `m/u` training — §Further Research — but only where blocking recall is the binding constraint.)
3. Add a **CI gate** locking F1 on a held-out labelled set.
4. **Gate go/no-go on internal corporate customer data** — the NC-voter run is only a public-data sanity check (pre-deduplicated; no DOB/email), so the production decision uses the org's own labelled data.

Part 3 · Line Engineers

Orientation — where the knobs are

What the code does

```
Blocker    : embed query → FAISS top-k candidates (cosine)
Linker     : Splink comparisons → per-candidate match probability
Store      : add / update / delete / save / load (no re-embed on load)
```

- Match probability = Bayesian combination of per-field weights $w = \log_2(m/u)$, a match prior, thresholded at τ .
- In-memory index today; `IndexingStrategy` / `VectorDatabase` is the seam for a vector DB later.

Scorecard to remember

- Confusion matrix (5k refs / 15k queries): **F1 99.46%**, recall 99.41%, precision 99.51%.
- Blocking recall: **99.88%** @k=20 (99.95% compact).
- **7.48 ms/query** (embed + block + batched Splink).
- NC-voter (real, mutated): **F1 92–94%**; blocking is the binding constraint.

Baseline engineering numbers — config, hardware, and data shape all move them.

Reproducing experiments

Run from repo root; scripts have `--help` and deterministic seeds (default 42).

Paper section	Script
§7 evaluation	<code>scripts/experiment_section7_eval.py</code>
§8 confusion matrix	<code>scripts/experiment_confusion_matrix.py</code>
§8.1 m/u calibration	<code>scripts/experiment_mu_calibration.py</code>
§8.2 duplicate benchmark	<code>scripts/experiment_duplicate_benchmark.py</code>
§8.3 F1 sweep	<code>scripts/experiment_f1_sweep.py</code>
calibration paradox	<code>scripts/experiment_mu_tau_interaction.py</code>
joint $\tau \times$ prior surface	<code>scripts/experiment_mu_prior_tau_surface.py</code>
NC-voter replication	<code>scripts/ncvoter/*</code> + <code>extract_ncvoter.py</code>

For **your** data: population-based scripts accept `--input-records FILE`.

Gotchas — tuning & thresholds

- **Don't "just retune" m/u .** It looks worse (prior/ τ coupling). Anchor the prior and re-validate prior + τ + weights together (Appendix — Tuning the Match Prior).
- **τ encodes business cost.** $\tau^* = C_{FP} / (C_{FP} + C_{FN})$; exposed via `--threshold` / `--tau` and the F1 sweep. Raise τ for fewer wrong links, lower it for recall-first (Appendix — Deriving τ).
- **Tune the prior properly.** Anchor λ , fit with the prior pinned, tune τ jointly, watch the score-shift guard (Appendix — Tuning the Match Prior).

Gotchas — blocking & data

- **Blocking recall is a hard ceiling.** If recall is low, raise `k` or serialization, not the embedding model.
- **Address weakening didn't help** here — test, don't assume.
- **Real data needs a mutation/duplicate model** to score (`scripts/ncvoter/ncvoter_util.py`).
- **NC voter has no full DOB and no email** — those comparisons are weak there.

Where the details live

Whitepaper: `docs/entity_resolution_whitepaper.pdf` (source: `.tex`).

- **§3 Two-Stage** — blocking/linkage contract and costs.
- **§7 Evaluation** — recall, thresholds, latency, ablations.
- **§8, 8.1, 8.2, 8.3** — headline results and the "don't retune alone" evidence.
- **§9 NC-voter** — real schema, mutation model, `k` scaling.
- **Appendix: Deriving τ** — cost / F1 / GMM / transitivity methods.
- **Appendix: Tuning the Match Prior** — anchor `$\lambda$` , then tune `$(\lambda, \tau)$` jointly (the calibration-paradox fix).
- **Use of AI** — disclosure and verification.

Checklist before you leave

1. Run `scripts/experiment_confusion_matrix.py --count 5000` and read the JSON.
 2. Reproduce the paradox: `scripts/experiment_mu_tau_interaction.py --base-count 5000`.
 3. Swap `--input-records <your file>` into a population-based experiment.
 4. Find where **blocking recall** caps your F1 — then plan linkage work.
- That is the fastest way to make these results your own.

Thank you

FAISS + Splink — one pipeline, three audiences

Bring a labelled sample of your real duplicates.